

Gamma hedging sensitivity impact on Zero-Days-to-Expiration options trading strategies profitability

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This paper is devoted to the sensitivity analysis of Zero-Days-to-Expiration options (0DTE options) profitability to the gamma hedging. Trading in Zero-Days-to-Expiration options, so called 0DTEs, have exploded in popularity in recent years – rising from 5% of SPX options volume in 2016 to over 40% in 2023.

0DTE options are option contracts that exist for a single trading session and expire on the same day that they are traded. A 0DTE option could be a longer-term option that has reached the last day of its lifecycle, or it could be a specific option that is listed only for a single day. Interest in 0DTE options, and other short-dated options (5DTE or less) has grown significantly in recent years. For example, the NYSE reported that 56% of all retail options volume is now in options with five or fewer days to expiration, as compared to about 35% in November 2019.

As volumes have increased, so have concerns around the market impact of these products. Specifically, the fear is that market makers hedging these options could become outsized relative to the underlying S&P market, and therefore option “gamma hedging” may be exerting undue influence on the market.

Most of the concerns around SPX 0DTE options arise because of their massive volume - averaging over 1,23 mln. contracts (\$500 bln. notional) in 2023 year. While the numbers may sound big, it's important to emphasize that volume doesn't equate to risk. High notional doesn't necessarily mean market makers on the other side of the trade will need to do a lot of hedging. What matters is the balance of the volume between «buys» and «sells», not the total size of the volume. To simplify, if 100 000 contracts trade on a particular strike and 50 000 was customers buying and the other 50 000 was customers selling, then despite \$50 bln. notional trading on that line, the amount market makers need to buy/sell in S&P futures to hedge is actual zero. Because the flow is perfectly balanced, market makers are left with net zero gamma risk despite the large volume in the 0DTE options [1].

Analysis of gamma hedging by market makers

Gamma hedging is an options trading strategy used by traders in the event of rare occurrences, and it has low probability, high impact in nature. This strategy is usually used with options that are close to their expiration date, while the underlying security on which the option is based is highly volatile. Experienced traders use this approach when they wish to reduce their risk exposure on either side of the trade. Options generally become more volatile as their respective expiration date nears, and gamma hedging is the answer to minimizing price risk in such event.

Mechanics of the Gamma hedging trade. When the gamma exposure is negative (positive), delta decreases (increases) when the price of the underlying asset increases. Hence market makers that engage in delta-hedging strategies are required to buy (sell) the underlying more aggressively after an increase in the underlying's price. While a market maker has sold many call contracts in their current portfolio, gamma hedging revolves around adding diverging option positions (and sometimes stocks) to said portfolio. So, a market maker offsets this upside risk with gamma hedge by adding call options or the underlying shares to the portfolio. As prices rise, the market maker must add positive exposure thereby adding “fuel to the fire” by buying more. This additional buying can become a tsunami of positive price action. However, when the price action turns negative, the market maker must reverse course and reduce exposure quickly, thus adding selling pressure.

Hedging short gamma exposure requires trading in the direction of price movements, thereby creating price momentum. Market intraday momentum is bound with gamma hedging demand from market makers.

Once we have an accurate measure of market maker's net positioning for each strike, we can calculate an aggregate net gamma number. That number will tell us how much market makers' delta exposure in 0DTE options will change in response to a 1% move in the SPX index – and as a result, how much potential buying or selling in S&P futures they may need to do in order to stay hedged. The bigger the number of each asset required, the higher the potential impact.

Despite the huge notional volume that is being traded in SPX 0DTE options on a daily basis, actual net exposure for market makers is fairly minimal, with average net gamma ranging from 0,04% to 0,17% of the daily S&P futures liquidity. There is also no discernible market impact from 0DTE option trading, with SPX

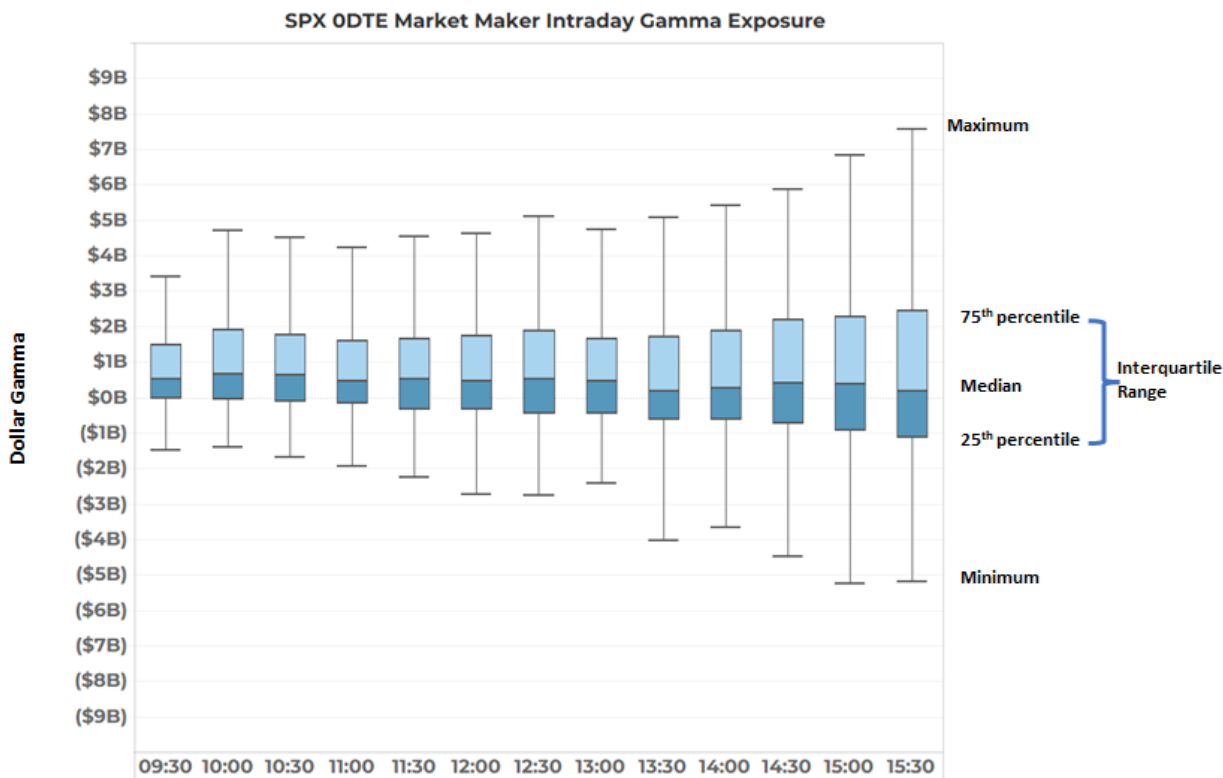
index intraday volatility and price patterns in line with historical averages. This is largely due to the balanced nature of 0DTE trading, with both institutional and retail investors finding a diverse range of use cases for the product.

Given the ultra low bid-offer spreads inherent to very low premium 0DTE, substantially less friction is present for active trading of 0DTE option spread positions than in the past, thereby reducing hedging costs. There is evidence to suggest that this fact is driving some of the 0DTE volume increase by pro-tail traders using automated spreading engines. Moreover, several traders are now more easily open and close multiple positions in 0DTE.

The chart 1 below shows the range of market maker net gamma positioning throughout the day, taken in 30-min increments over the past year. The boxes tell the interquartile range from the 25th to 75th percentile of the data (i.e. the middle 50% of the observations over the past year) with the median represented by the horizontal bar in the middle of the box. The “whiskers” extend to the min/max of 1,5x the interquartile range (anything beyond that would be considered outliers) [2].

The range of market maker net gamma positioning throughout the day

Chart 1



When the gamma exposure is negative (positive), delta decreases (increases) when the price of the underlying asset increases. Hence market makers that engage in gamma-hedging strategies are required to buy (sell) the underlying more aggressively after an increase in the underlying’s price. This results into additional positive (negative) price pressure, which increases (decreases) the magnitude of the initial price movement. Thus, the initial price movement is dampened (reinforced) when the net gamma exposure is positive (negative). Hence, the relation between net gamma exposure and volatility is expected to be negative. This relationship also implies that risk-averse investors tend to be averse towards negative net gamma exposure, and demand a compensation to hold such stocks. Options with positive net gamma exposure are considered as safer assets. In that case, investors are willing to pay higher prices, and accept lower expected returns. Investors will only pay high prices for options with positive gamma hedging pressure in the past with the expectation that such pressure will persist over time. Net gamma exposures also negatively predict extreme returns, consistent with the idea that positive gamma hedging acts as a volatility dampener.

0DTE OPTIONS TRADING STRATEGIES

The difference between a 0DTE option and a regular weekly or monthly option is the reduced time until expiration. Theoretically, that means the full spectrum of options trading strategies may be applied to zero-day options. Practically, however, 0DTE options aren't being heavily utilized for sophisticated trading approaches. Instead, most market participants are using 0DTE options to play one-day moves in the market using index ETFs like the SPDR S&P 500 ETF Trust (SPY), which now offers 0DTE options for every single trading day on the calendar.

In most cases, these options are being used to deploy concentrated directional risk using naked calls or naked puts (as an example, directional options strategies), for example, loading up on out-of-the-money (OTM) calls to try and ride a sharp upside rally in the S&P 500. Traders must first ascertain what posture they want to take in the market at the portfolio level. This could range from very bullish to slightly bullish, or very bearish to slightly bearish, or even market neutral (i.e. as flat as possible from a directional perspective). Depending on the result of this analysis, the next step is to determine how much aggregate directional risk exists in the current portfolio of 0DTE options. After identifying the degree and type of directional risk required to deploy in the market, traders can then consider what adjustments might be needed in their portfolio. For example, if a trader has decided to deploy a slightly bearish market posture, but is currently positioned market neutral, they might decide to add additional short deltas to the portfolio of 0DTE options as shown on the Table 2.

Adding short deltas to the 0DTE options portfolio

Table 2

Ways	For adding short deltas trader can use:
New positions	1) Long Put Spreads 2) Short Call Spreads 3) Short Stocks 4) Short Futures
Closes/Adjustments	1) Close Bullish Trades 2) Roll Calls In (down)

As shown on the Table 2, traders can use a variety of methods to address aggregate directional exposure in their portfolios. And while the exact method / tactic will vary, the most important thing is that this risk dynamics is actively addressed. Maintaining gamma neutrality is a crucial aspect of 0DTE options trading, as it allows traders to hedge against directional risk and focus on other factors such as volatility (Vega) and time decay (Theta).

Key 0DTE trading strategies

- **Neutral options strategies:**

In the face of market uncertainty, two 0DTE trading strategies offer a haven of relative calm:

- 1) *iron butterfly*;
- 2) *iron condor*;

Iron butterfly and iron condor options combine calls and puts at various strike prices, creating a profit zone where small price movements in either direction can generate a consistent profit. It is possible to imagine them as protective shields, capturing gains regardless of the underlying asset's price direction. Iron butterflies offer a defined risk and reward profile, while iron condors provide a more structured approach for those seeking a predictable trading approach.

- 3) *straddle*;
- 4) *strangle*.

Straddles and strangles involve buying both a call and a put at the same or different strike prices, capitalizing on significant price swings regardless of direction.

• **Non-directional options strategies:**

Options traders employ a variety of approaches here, but short-term selling 0DTE options is particularly compelling also known as ‘zeroing in on Gamma”, this non-directional option strategy involves dynamically selling premium to capture swelling time decay as expiration nears. The concept is straightforward - volatile 0DTE premiums deflate rapidly during the session and thereby allowing sellers to buy options back cheaper than they received in credit. Rather than predicting uptrends / downtrends, the focus becomes targeting extrinsic value melt.

• **Directional options strategies:**

- *naked Calls/Puts*: this strategy involves selling uncovered calls or puts, aiming to profit from a strong directional move in the underlying asset;
- *debit Spreads*: this strategy involves buying a call and selling a higher-strike call or vice versa, aiming for a smaller but more controlled profit from a directional move.

Table 3 shows that targeting iron butterflies with small profit targets and stop losses led to successful trading.

Table 3

Gamma hedging impact and probability of profit of 0DTE trading strategies

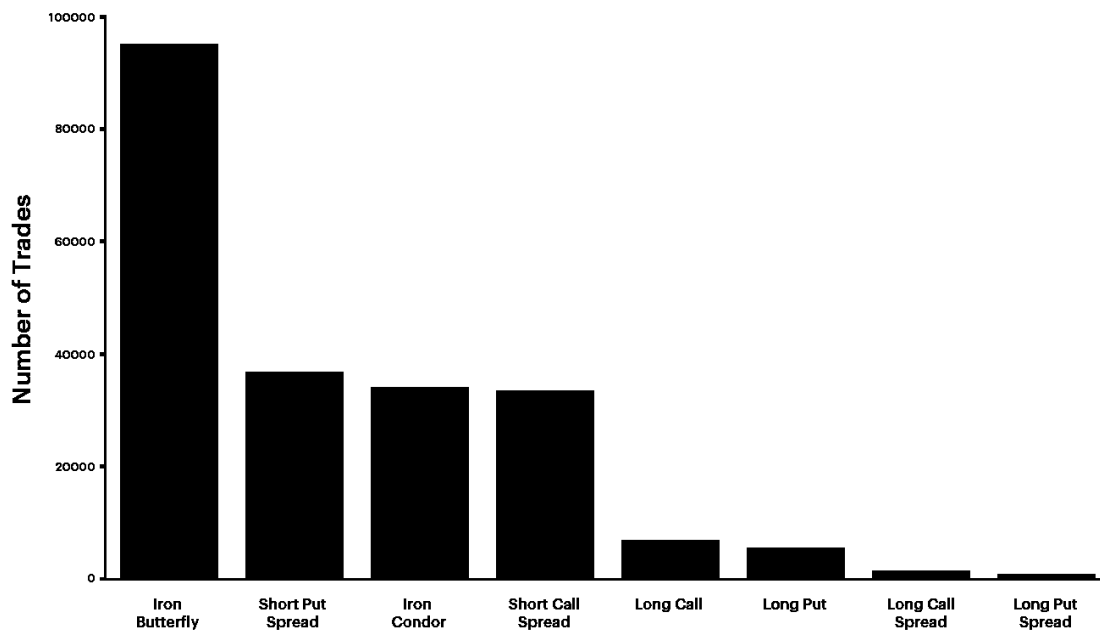
Trading Strategy	Probability of profit (avg.)	Gamma hedging techniques and it's impact	Gamma behavior
Iron Butterfly	20-30%	Traders can employ gamma hedging strategies to protect their positions from adverse price movements. For example, if the underlying asset's price starts to move towards one of the breakeven points of the iron butterfly, the gamma of the options will change, altering the position's delta. To hedge against this risk, traders can adjust their positions by buying or selling additional options to maintain a neutral delta. This allows them to protect their profits and limit potential losses.	• Positive gamma when underlying price moves outside the wings of the iron butterfly; • Negative gamma when the underlying price is in between the wings of the iron butterfly.
Iron condor	60-70%	Traders can buy or sell options with high gamma to hedge. For example, if the delta of the call options in the iron condor is increasing, traders can buy call options with high gamma to offset the increase in delta.	• If the P&L changes for the worst, there may not be enough time left in the trade to recover that loss. When iron condor gets closer to expiration, theta increases, and correspondingly, the magnitude of gamma increases, making the trade more sensitive to price movements. • If a trader finds that gamma is causing the delta to change too much, having to make too many adjustments, they may prefer to alter the trade to decrease gamma – either by narrowing the widths of the spreads.
Long straddle	30-40%	Gamma hedging technique: add gamma to the position's delta when	Positive gamma

		the underlying stock price increases, and subtract gamma from the position's delta when the underlying stock price falls.	
Short straddle	30-40%	Gamma hedging techniques that traders can use: • <i>Dynamic delta hedging.</i> This technique involves continuously adjusting the delta of the options portfolio to maintain a neutral position. This means that the trader buys or sells shares of the underlying stock to offset any changes in the options' delta. For example, if the delta of the 0DTE options portfolio becomes too positive, the trader would sell shares of the underlying stock to reduce the delta to a more neutral level. • <i>Gamma scalping.</i> This technique involves buying and selling shares of the underlying stock to maintain a neutral gamma position. Traders need to buy or sell shares of the underlying stock to offset any changes in the options' gamma. For example, if the gamma of options portfolio becomes too positive, the trader would buy shares of the underlying stock to increase the gamma to a more neutral level. • <i>Vega hedging.</i> This technique involves adjusting the options portfolio's vega to maintain a neutral position. Traders can use this technique to protect themselves from unexpected changes in implied volatility. For example, if the implied volatility of options portfolio increases, the trader would sell options to reduce the vega to a more neutral level.	Negative gamma
Long strangle	40-55%	Long and short strangles are not typically hedged upon position deployment, because both are theoretically delta neutral.	Positive gamma
Short strangle	50-55%	However, the position holder might elect to gamma hedge a long or short strangle if the underlying stock makes a big move in either direction.	Negative gamma
Long Naked		To hedge gamma of a long call, an investor may purchase a put with	Positive gamma

Call	50-60%	the same strike price and expiration date, thereby creating a long straddle.	
Short Naked Call	70-80%	To hedge gamma of a short call, an investor may sell a put with the same strike price and expiration date, thereby creating a short straddle	Negative gamma
Long Naked Put	30-40%	To hedge gamma of a long put, an investor may purchase call with the same strike price and expiration date, thereby creating a long straddle.	Positive gamma
Short Naked Put	70-80%	To hedge gamma of a short put, an investor may sell a call with the same strike price and expiration date, thereby creating a short straddle	Negative gamma
Debit spread (Bull-call spread)	65-70%	Gamma changes very little (no need for gamma hedging)	Near-zero gamma

The statistics shown on the second column of Table 3 is confirmed by the chart 4.

Chart 4



Gamma behavior of 0DTE options is shown on the Table 5.

Gamma behavior of 0DTE options

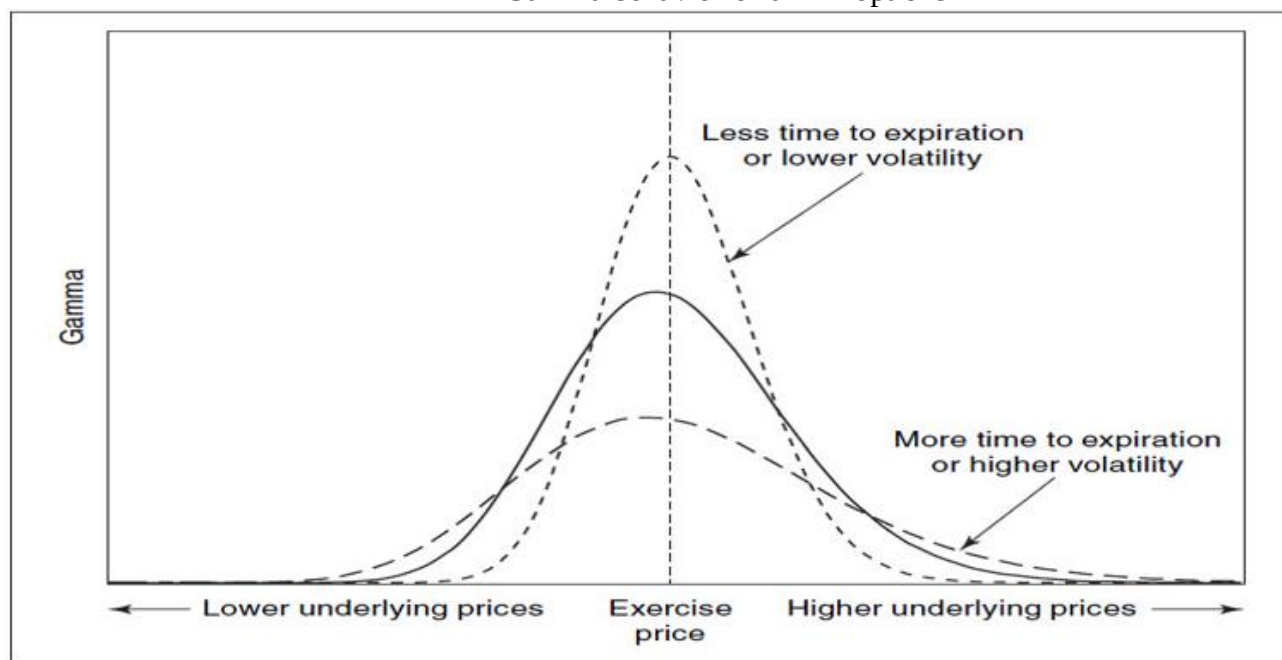


Table 5 shows that the impact of shortening the time to expiration to one day or less dramatically increases Theta to reflect the premium over the few remaining hours of the life of the 0DTE options. In cases of at-the-money or near-the-money strikes gamma increases as a reflection of the potentially significant Delta change caused by small moves in the underlying asset price. Gamma decreases at out-of-the-money strikes as probability of achieving in-the-money state become lower along with the correspondingly lowering Delta and Theta.

0DTE options with high net gamma exposure systematically underperform options with low net gamma exposure. This effect is distinct from other well-known return predictors, and survives many robustness checks. Options with low net gamma exposure negatively predict future realized volatility, and argue that investors command a risk premium to hold low net gamma exposure options, which are riskier.

Trading features of 0DTE

Most 0DTE traders are aware of assignment risk trading so close to expiration, so they are proactive in closing trades manually before the end of the trading day. Traders should keep in mind that the costs and risks associated with 0DTE options can be higher as compared to weekly and monthly options. Participants in the options market are encouraged to review that information—covering the potential benefits and drawbacks of 0DTE options. The lack of extrinsic value is one of the reasons why 0DTE options are sensitive to underlying price changes, since they can go from being worthless to being worth on a small stock price move. As a result, traders who speculate correctly can see quick profits if they choose the correct direction. However, traders who speculate incorrectly can experience rapid losses, whether long or short. Gamma drives the pace of trading 0DTE options and options near expiration because of their potential to yield fast profits, but it is bound with risk of loss as traders have less time to take right decision.

Key inferences:

- Iron condor trading strategy for 0DTE options is the most profitable among others and gamma hedging impacts moderate on it's profitability;
- Despite the huge notional volume that is being traded in SPX 0DTE options on a daily basis, actual net gamma exposure for market makers is fairly minimal comparing with other types of options;
- Gamma hedging is more crucial in the states such as at-the money or near-the-money strikes as gamma at these states increases sharply;
- Directional risk become cornerstone during 0DTE trading;
- Directional options strategies demonstrate best profit results after neutral strategies (especially Naked Short Call and Naked Short Put).

References:

[1] J.Boss, 2023, «The 0DTE trader: A guide to profitable options trading in no time», p. 139–143;

[2] B.Cui; 2023, «The 0DTE options handbook: profit from zero-days-to-expiration options in any market condition», p. 85–89.